Example 1 - rivt Doc

Attn: **user example**

project: **0001**

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1. Summary

This rivt file example calculates the maximum stress and deflection in a simply supported, uniformly loaded beam. It also serves as an annotated example of a rivt doc with multiple sections that is not part of a report.

The example illustrates the use of some of the most common API functions, commands and tags. Further details are provided in the rivt user manual.

The file may be formatted as a text, PDF or HTML doc by changing the type parameter in the PUBLISH command in the Doc-API *(rv.D)* at the end of each rivt file. Published files are found in the sub-folders of the *_published* folder.

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2. Load Combinations

Table 1: ASCE 7-05 Load Effects (saved as csv in _stored folder)

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

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3. Loads and Geometry

Successive value definitions are formatted as a table. Variable values are defined with the define operator. The line tag [T] labels and numbers the table.

Table 2: Define Unit Loads

variable	value	[value]	description
D_1	3.80 psf	0.18 kPA	joists DL
D_2	2.10 psf	0.10 kPA	plywood DL
D_3	10.00 psf	0.48 kPA	partitions DL
D_4	3.00 klf	43.78 kN_m	fixed machinery DL
L_1	40.00 psf	1.92 kPA	ASCE7-O5 LL
b_1	10.00 inch	254.00 mm	beam width
h_1	18.00 inch	457.20 mm	beam depth
E_1	29000.00 ksi	199947.96 MPA	modulus of elasticity
Fb_1	20000.00 lb_in2	137.90 MPA	allowable stress

The VALTABLE command reads variable values from a file in the src folder. The description is used as the table title. The range specifies the starting and ending line to be read from the file (0:0 means all lines).

Table 3: Beam Geometry from file: **src/beam1.csv**

variable	value	[value]	description
spc_1	2.00 ft	0.61 m	beam spacing
spn_1	16.00 ft	4.88 m	beam span

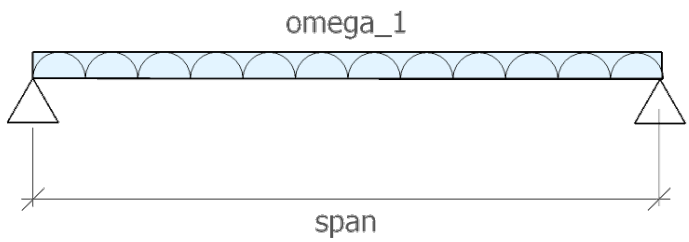

Fig. 1 - Beam Diagram

Uniform Distributed Loads

Eq. 1: Dead load [ASCE7-05 2.3.2]

```
dl₁ = 1.2·(D₄ + spc₁·(D₁ + D₂ + D₃))

dl₁ = 3.64 klf     [dl₁] = 53.09 kN_m   | Dead load [ASCE7-05 2.3.2]

spc₁          D₁         D₂          D₄                 D₃
────────────  ─────────  ──────────  ─────────────────  ────────────
2.00 ft       3.80 psf   2.10 psf    3.00 klf           10.00 psf
─────         ─────      ─────       ─────              ─────
beam spacing  joists DL  plywood DL  fixed machinery DL partitions DL
────────────  ─────────  ──────────  ─────────────────  ────────────
```

Eq. 2: Live load [ASCE7-05 2.3.2]

```
ll₁ = 1.6·L₁·spc₁

ll₁ = 0.13 klf     [ll₁] = 1.87 kN_m   | Live load [ASCE7-05 2.3.2]

spc₁          L₁
────────────  ──────────
2.00 ft       40.00 psf
─────         ─────
beam spacing  ASCE7-05 LL
────────────  ──────────
```

Eq. 3: Total load [ASCE7-05 2.3.2]

```
ω₁ = dl₁ + ll₁

ω₁ = 3.77 klf     [ω₁] = 54.96 kN_m   | Total load [ASCE7-05 2.3.2]

dl₁                   ll₁
────────────────────  ────────────────────
3.64 klf              128.00 ft·psf
─────                 ─────
Dead load [ASCE7-05   Live load [ASCE7-05
2.3.2]                2.3.2]
────────────────────  ────────────────────
```

---

4. Beam Response

The following lines import the beam geometry from an external file, calculate section properties from imported functions and calculate the maximum moment, bending stress and mid-span deflection.

Table 4: Beam functions from file: **src/sectprop.py**

Function	Docstring
rectsect(b, d)	section modulus of rectangle
rectinertia(b, d)	moment of inertia of rectangle
midspan_delta(ln, w, e, i)	mid-span deflection of simply supported beam with UDL

Eq. 4: rectangle - S (sectprop.py)

```
section₁ = rectsect(b₁, h₁)

section₁ = 540.00 in3     [section₁] = 8849.01 cm3   | rectangle - S (sectprop.py)

b₁           h₁
──────────   ──────────
10.00 inch   18.00 inch
─────        ─────
beam width   beam depth
──────────   ──────────
```

Eq. 5: rectangle - I (sectprop.py)

```
inertia₁ = rectinertia(b₁, h₁)

inertia₁ = 4860.0 in4     [inertia₁] = 202288.5 cm4   | rectangle - I (sectprop.py)

b₁           h₁
──────────   ──────────
10.0 inch    18.0 inch
─────        ─────
```

beam width	beam depth

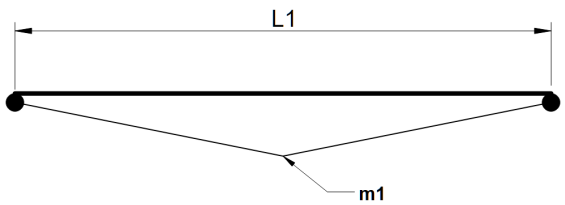

Fig. 2 - Moment diagram

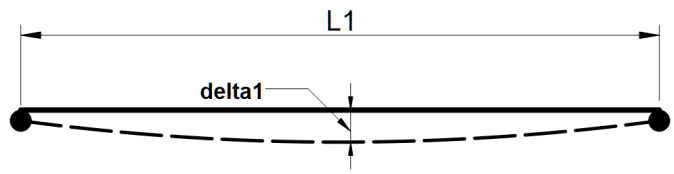

Fig. 3 - Deflection diagram

Eq.6: Maximum bending stress formula

$$\sigma_1 = \frac{M_1}{S_1}$$

Eq. 7: Mid-span UDL moment

$$m_1 = \frac{\omega_1 \cdot spn_1^2}{8}$$

$m_1 = 120.52\ ftkip$ $[m_1] = 163.40\ mkN$ | Mid-span UDL moment

spn_1	ω_1
16.00 ft	3.77 klf
beam span -	Total load [ASCE7-05 2.3.2]

Eq. 8: Bending stress

$$fb_1 = \frac{m_1}{section_1}$$

```
fb₁ = 2678.2 lb_in2     [fb₁] = 18.5 MPA   | Bending stress

m₁                    section₁
____________________  _____________
120.5 ft2·klf         540.0 inch3
_____                 _____
Mid-span UDL moment   rectangle - S
-                     (sectprop.py)
____________________  _____________
```

Eq. 9: Stress ratio

```
▮ fb₁ < Fb₁
▮
▮ fb₁       Fb₁        ratio     check    reference
▮ ________  _________  ________  _______  ____________
▮ 2.68 ksi  20.00 ksi  0.133908  OK       Stress ratio
▮ ________  _________  ________  _______  ____________
```

Eq. 10: mid-span deflection (sectprop.py)

```
δ₁ = midspan_δ(spn₁, ω₁, E₁, inertia₁)

δ₁ = 0.04 inch     [δ₁] = 1.00 mm   | mid-span deflection (sectprop.py)

spn₁       inertia₁       E₁                     ω₁
_________  _____________  _____________________  ____________________
16.00 ft   4860.00 inch4  29000.00 ksi           3.77 klf
_____      _____          _____                  _____
beam span  rectangle - I  modulus of elasticity  Total load [ASCE7-05
-          (sectprop.py)  -                      2.3.2]
_________  _____________  _____________________  ____________________
```